

The Decoupling Locus: a MOND Scale Derived from a Clock, and the Solar System Reduced to One Number

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Abstract

A cuscuton-clock sector with a projected-gradient scalar produces a cold clustering component and passes nine cosmological and solar-system gates as a dark component [3, 4, 5, 6], but produces no force law, because matter in that action is minimally coupled: $\delta S_{\text{matter}}/\delta\chi = 0$ identically. Adding a coupling is what would give MOND, and the obstruction was that the clock and the scalar mix, so a coupled matter source would drag the clock and generate preferred-frame effects. We report that the mixing vanishes *exactly* on the locus $W_0 \equiv U - 2\gamma\bar{q}^2\dot{\bar{q}} = 0$, which is a condition on coefficients the action already has. A first pass priced that locus at a cubic coupling eight orders too large, scaling as $1/w$; that pricing held the clock coefficient U fixed, and U is not free. With $\rho = U/m_{\text{rel}}$ imposed the factor of w cancels identically: what the locus fixes is the dimensionless ratio $\gamma\bar{q}^3 H/\rho = -1/[6(s_0 - 1)]$, of magnitude at most $1/6$, and the operator's weight against gravity is $\Omega_{\text{sec}}/[6(s_0 - 1)] \leq 0.044$. The earlier verdict is withdrawn. The locus also *forces* two previously free quantities: positivity of the sector's remaining energy requires $s_0 \geq 2$, hence $m_{\text{rel}} \leq w \lesssim 10^{-4}$. Extending the action by the $Y^{3/2}$ operator identified as missing in [7] and a leading-order matter coupling, the quasi-static gradient sector yields the deep-MOND scaling outright ($d \log g/d \log r = -1$, $d \log g/d \log M = +1/2$) and fixes the acceleration scale, $a_0 = \lambda^3/(12\pi G\beta s_0)$, mass-independent and falling with the clock rate. The single conflict — the branch wants the linear gradient coefficient large, deep MOND wants it small, and it is the same coefficient — resolves into one inequality, $C \equiv 2k\bar{q}^2/U \gtrsim 10^3$, whose derived exponents cancel exactly, so C is a constant of the motion. Finally, coupling matter *disformally* along the clock's own unit timelike direction shifts both weak-field potentials equally, giving $\gamma_{\text{PPN}} = 1$ exactly with no added vector field; boosted, the same term gives $\alpha_1 = 8f_s$, five orders over the bound unless local matter drags the clock to within 4.6 m s^{-1} . That alignment, and κ , are what remain open. Ten new Lean 4 certificates; the file stands at 124 theorems, zero **sorry**.

1 Position, and three corrections to the previous paper

The action [2] is, with $c = 1$ and signature $(-+++)$,

$$S = \int d^4x \sqrt{-g} \left[\frac{M^2}{2} (R - 2\Lambda) + P(X, \tau) - V(\tau) + sW(Y, \tau) + \gamma X \square \chi \right] + S_r + S_b, \quad (1)$$

with $s = \sqrt{-\nabla_\mu \tau \nabla^\mu \tau}$, $n_\mu = -\partial_\mu \tau/s$, $Q = n^\mu \partial_\mu \chi$, $Y = (g^{\mu\nu} + n^\mu n^\nu) \partial_\mu \chi \partial_\nu \chi$, $X = Q^2 - Y$. An external audit of [7] found three errors, each reproduced independently before being accepted, and they are recorded here because two of them were load-bearing.

(i) *The health condition had the wrong sign.* We stated $4d\ell > U$. Checking the same reduction against the canonical scalar $P(X) = X/2$, which gives $F_Y = -\frac{1}{2}$, shows a healthy field has *negative* F_Y . The correct condition is $\boxed{U > 4d\ell}$. Every verdict resting on the reversed inequality is withdrawn, including a proposed coefficient window.

(ii) *The interpolating exponent was measured against the wrong acceleration.* The reported slope 0.470 compared the kernel with the Newtonian acceleration $x = g_N/a_0$ rather than the true one $y = x\nu(x)$. Converted, the slope is 0.997. The qualitative conclusion of [7] — that MOND requires a linear μ in y , which the unextended action does not supply — survives with the number corrected.

(iii) *The no-force-law verdict was right for a different reason.* [7] excluded a force law on the exponents of the two gradient operators. The stronger and simpler fact is that matter in (1) enters only through S_r and S_b , both minimally coupled to $g_{\mu\nu}$: $\delta S_{\text{matter}}/\delta\chi = 0$ identically. The action modifies *no* force law, and the rotation-curve exponents of [7] are conditional on a coupling that is not present. Correspondingly, a structural argument that a vanishing background gradient removes the clock–scalar mixing is also withdrawn: the exact quadratic action carries a cross term $-2qW_Y$ and a clock response $\sigma/\pi = 2qsW_Y/(2q^2W_Y - W_0)$, both of which survive $Y \rightarrow 0$. The preferred-frame gate returns to open, and that is the gate this paper addresses.

2 The locus on which the clock stops listening

The clock response above vanishes identically, together with the entire W_Y contribution to the eliminated scalar coefficient, when

$$W_0 \equiv U - 2\gamma\bar{q}^2\dot{\bar{q}} = 0, \quad (2)$$

at which point that coefficient reduces to $2P_X$ alone. Since W_0 is built from quantities (1) already contains, this is a locus inside the existing parameter space, not a modification. It has one immediate cost: with $W_0 = 0$ every surviving term of the clock equation carries the same factor of the clock rate, which therefore cancels. That equation no longer determines s_0 but constrains the coefficient functions; s_0 remains fixed by conservation through $s_0 - 1 = w/m_{\text{rel}}$.

3 What the locus costs, measured in units that cannot be inflated

The stress tensor, derived rather than quoted. Write $S = \int dt Na^3 L_m$ in minisuperspace with the lapse restored. Then $\rho = -a^{-3}\partial_N(Na^3 L_m)|_{N=1}$ and $p = (3a^2)^{-1}[\partial_a - d_t\partial_{\dot{a}}](Na^3 L_m)|_{N=1}$. Calibrating on $L_m = F(Q)$ reproduces the textbook pair $QF_Q - F$ and F . Applied to the cubic operator, whose minisuperspace action is $-2\gamma a^2 \dot{a} Q^3$ after an integration by parts that differs from the unintegrated form by an explicit total derivative,

$$\rho_3 = -6\gamma H \bar{q}^3, \quad p_3 = 2\gamma \bar{q}^2 \dot{\bar{q}}, \quad (3)$$

so that $W_0 = U - p_3$: *the decoupling condition says the cubic operator's pressure equals the clock coefficient.* On the derived family $\bar{q} \propto a^{-3w}$ one finds $p_3 = w\rho_3$ identically, so the operator carries the sector's own equation of state and switching it on does not disturb the background scaling.

The correction. Using the family's $\rho = U/m_{\text{rel}}$ and $p = U(s_0 - 1)$ [5], the cubic operator's share of the sector density is

$$\frac{\rho_3}{\rho} = \frac{1}{s_0 - 1}, \quad (4)$$

free of both U and w . A first pass had reported $|\gamma| = U/(6|w|\bar{q}^3 H)$ and concluded, holding U fixed, that the acoustic bound $w \lesssim 10^{-4}$ forces $|\gamma| \sim 650$ against the 10^{-6} used throughout. But $U = m_{\text{rel}}\rho$ with $m_{\text{rel}} = w/(s_0 - 1)$, and substituting, $\partial|\gamma|/\partial w = 0$ exactly. The divergence was an artifact of units in which the sector density had been set to $1/m_{\text{rel}}$, which is large for the same reason w is small. What the locus fixes is a ratio,

$$\frac{\gamma\bar{q}^3 H}{\rho} = -\frac{1}{6(s_0 - 1)}, \quad (5)$$

of magnitude at most $1/6$; today $|\gamma|\bar{q}^3 = 1.13 \times 10^{21} \text{ eV}^3$, a cube root of 10.4 MeV . Comparing the cubic operator's share of the momentum conjugate to the scale factor with the Einstein–Hilbert term's, from the same action,

$$\left| \frac{\pi_3}{\pi_{\text{grav}}} \right| = \frac{\Omega_{\text{sec}}}{6(s_0 - 1)} \leq 0.044, \quad (6)$$

a percent-level correction. Every result derived at $\gamma \rightarrow 0$ therefore survives on this locus to about four percent, and the earlier verdict that they were invalidated is withdrawn.

Two free quantities become forced. The non-cubic density is $\rho[1 - 1/(s_0 - 1)]$, which is non-negative only for

$$s_0 \geq 2 \quad \Longleftrightarrow \quad m_{\text{rel}} \leq w \lesssim 10^{-4}. \quad (7)$$

The clock must run at least twice proper time, and the margin must sit *below* the equation of state rather than above it. At the boundary $s_0 = 2$ the cubic operator supplies the entire dark sector density on its own. Both statements are Lean-certified.

4 The force law on the locus

Extend W by the $Y^{3/2}$ operator [7] identified as the missing ingredient, and couple matter at leading order, $\lambda\varphi\rho_m$. In the quasi-static limit the gradient equation is $\nabla \cdot (2sW_Y \nabla \varphi) = -\lambda\rho$, so $2sW_Y$ plays the role of the interpolating function. With $W = \beta Y^{3/2}$ the spherical solution gives an acceleration on matter with

$$\frac{d \log g}{d \log r} = -1, \quad \frac{d \log g}{d \log M} = +\frac{1}{2}, \quad (8)$$

the deep-MOND scaling, obtained from the action rather than assumed. Matching to $\sqrt{GMa_0}/r$ fixes the scale in terms of the action's own coefficients,

$$\boxed{a_0 = \frac{\lambda^3}{12\pi G\beta s_0}} \quad \text{with} \quad \frac{\partial a_0}{\partial M} = 0, \quad (9)$$

so it is a constant of the theory rather than an object-by-object fit, and it *falls as the clock runs faster*. On the canonical footing $\lambda^3/(\beta s_0) = 2.3555 \times 10^{-19}$ in SI units and on the alternative footing 2.8378×10^{-19} , the ratio being the footing fork 1.204777 itself.

The one conflict, and its window. The linear term d in W_Y appears twice: as the floor $\mu_{\text{floor}} = d/k$ it places under the interpolating function, k being W_Y at the highest acceleration probed, and as the closure's $\mu = 2d\bar{q}^2/U$ that sets the clock margin. They are the same d , related by $\mu = C \mu_{\text{floor}}$ with $C = 2k\bar{q}^2/U$. The locus needs $\mu \geq 1 - w$; the rotation-curve data need $\mu_{\text{floor}} \leq 10^{-3}$ for the

radial acceleration relation to survive at the ten-percent level down to $g/a_0 = 10^{-2}$. Both hold exactly when

$$C = \frac{2k\bar{q}^2}{U} \gtrsim 10^3. \quad (10)$$

On the derived family the exponents cancel, $-3(1-w) - 6w + 3(1+w) = 0$, so C is a constant of the motion: imposed once, it holds for all time. Missing the window does not kill the construction but buys it with a cancellation among the sector's densities of degree m_{rel}/w , which reaches 10^4 at $C = 1$ — the same $1/w$ the first pass saw, relocated from the coupling to the cancellation, where it belongs.

5 The solar system, reduced to one number

A *conformal* matter coupling shifts the time potential and not the space one and fails light bending outright. The standard cure is a disformal coupling along a unit timelike vector, which TeVeS [10] and AeST [11] must add as a separate dynamical field. Action (1) already has one: the clock's gradient supplies n_μ , and the cusciton [9] propagates no scalar of its own. Expanding $\tilde{g} = (1 - 2\varphi)g - 4\varphi n \otimes n$ in the weak field,

$$\tilde{\Phi} = \Phi + \varphi, \quad \tilde{\Psi} = \Psi + \varphi \quad \implies \quad \gamma_{\text{PPN}} = 1 \text{ exactly}, \quad (11)$$

however strong the scalar: the shift is common to both potentials and cancels from their difference. Light bending and Shapiro delay are general-relativistic.

Boosting is where it costs. For a system moving at w through the clock's frame the same term gives $\tilde{g}_{0i} = 4\varphi w_i$, first order in the velocity *and* first order in the scalar, which is the preferred-frame signature. Matching to the post-Newtonian form,

$$\alpha_1 = 8f_s, \quad (12)$$

f_s being the scalar's share of the local potential. At the share MOND requires at solar-system accelerations, $f_s \approx 1$, that is 8 against a bound of 10^{-4} : over by 8×10^4 . A clock sitting in the cosmic frame is excluded outright, by the same wall that closed the vector-based completions. The bound converts directly into an alignment requirement:

$$\boxed{v_{\text{residual}} \leq 4.6 \text{ m s}^{-1}} \quad (13)$$

between the clock's frame and the local matter frame, one part in 8×10^4 of the solar system's 370 km s^{-1} through the cosmic frame. *Whether the clock's own field equation delivers that is not computed here*, and an earlier lane measuring a cusciton clock as dragged at $3 \times 10^{-2} \Phi_N$ at Saturn [2] is the same question in a different parameterisation and was recorded there as a kill.

6 The board, and what is not claimed

Every gate the programme has posed now carries a computed number on this locus — the preferred-frame mixing (identically zero), the cubic backreaction (≤ 0.044), the clock rate (≥ 2) and margin ($\leq w$), the acoustic scale, the forest through criticality [4], lensing and S_8 (the sector is inert), and the force law and its scale — with two exceptions, which are stated as exceptions. The clock

alignment above is not computed. And $\kappa = \frac{1}{2}$ remains fitted and provably underivable by this class of actions [1]: that is a theorem of this programme, not a gap in this paper. **This is not a complete theory of gravity and nothing here should be read as one.**

Limits, in full. The elimination giving the mixing coefficient is an external audit's, used rather than re-derived. The positivity argument forcing $s_0 \geq 2$ assumes no piece of the sector carries negative energy; if one may, $s_0 < 2$ costs a cancellation of degree $(2 - s_0)/(s_0 - 1)$. The weight π_3/π_{grav} is a background measure; the perturbation kinetic matrix at $s_0 = 2$ has not been recomputed and no gate has been re-run numerically on the locus. The matter coupling is leading order, not a full conformal or disformal factor; the quasi-static reduction drops the clock's gradient response, legitimate only because the mixing vanishes here; k is defined, not derived; the interpolation used is the pure $Y^{3/2}$ limit, not ν_{RAR} in full, so the deep-MOND limit and the scale are established and the whole interpolating function is not. The post-Newtonian matching uses the standard normalisation, so the coefficient 8 is convention-dependent at the factor-of-two level; the five-orders conclusion is not.

7 Reproducibility

fable_independent_2026/ lanes L211_three_corrections.py, L212_decoupling_branch.py (commit 473860435), L213_branch_solved_and_the_board.py, L214_force_law_on_the_branch.py (commit 425a28355) and L215_solar_system_on_the_branch.py (commit bde5bfd2c), each with its committed output and results file. The Lean file fable_independent_2026/lean_2026/Mondlean.lean carries cubic_share_is_clock_rate, positivity_forces_clock_rate, margin_below_equation_of_state, cubic_coupling_is_equation_of_state_independent, mond_scale_mass_independent, branch_mond_window, derived_exponents_cancel, window_ratio_is_scale_invariant, disformal_gamma_is_one and preferred_frame_alignment_requirement; 124 theorems, exit 0, zero sorry, axioms $\subseteq \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$.

References

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